GWO — resultados CEC

Benchmark MAREA · paper OE1 · 71 funciones CEC2014/2017/2022 · 51 corridas por funcion

Protocolo

Valores = **error al optimo**, f(x) - f*. Menor es mejor; 0 es el optimo global. Dominio [-100,100] en todas las funciones. Poblacion N=80, iteraciones T=MaxFES/N. Presupuestos 5.000 / 50.000 / 500.000 evaluaciones. Las 30 funciones de CEC2014 y las 29 de CEC2017 corren en D=10 y D=50; las 12 de CEC2022 solo llegan a D=20, asi que en la columna D50 corren en **D=20** (marcado en la tabla). Todos los resultados vienen del mismo entorno del Mac mini (python 3.11.9, numpy 2.3.3, opfunu 1.0.4).

Resumen — posicion entre los 10 algoritmos (rango por media)

dimension	presupuesto	rango medio de Friedman	posicion de 10	funciones donde es el mejor
D10	5k FES	6.01	6º	0 de 71
D10	50k FES	7.06	8º	0 de 71
D10	500k FES	7.80	8º	0 de 71
D50	5k FES	3.65	3º	8 de 71
D50	50k FES	5.89	6º	1 de 71
D50	500k FES	7.49	8º	1 de 71

GWO — estadisticas por funcion, D = 10

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	10	7.75e+05	4.45e+07	1.09e+07	1.22e+07	8.71e+06	4.89e+05	3.85e+07	4.46e+06	6.56e+06	6.10e+06	1.82e+05	1.08e+07	3.54e+06	3.63e+06	2.75e+06
F22014	2014	10	6.28e+04	2.82e+08	3.35e+05	8.54e+06	4.25e+07	7.13e+02	2.33e+04	3.70e+03	5.81e+03	5.01e+03	9.63e+00	1.19e+04	8.00e+02	3.33e+03	4.23e+03
F32014	2014	10	2.19e+03	5.78e+04	2.03e+04	2.05e+04	1.29e+04	2.43e+02	1.79e+04	6.00e+03	7.72e+03	5.08e+03	1.87e+02	1.37e+04	1.62e+03	3.17e+03	3.20e+03
F42014	2014	10	6.26e+00	5.56e+01	3.59e+01	3.44e+01	8.31e+00	4.99e+00	3.86e+01	3.53e+01	3.23e+01	8.99e+00	2.94e-01	3.79e+01	3.50e+01	2.88e+01	1.15e+01
F52014	2014	10	2.03e+01	2.08e+01	2.06e+01	2.06e+01	1.14e-01	4.22e+00	2.05e+01	2.04e+01	2.01e+01	2.27e+00	3.84e-01	2.04e+01	2.03e+01	1.99e+01	2.79e+00
F62014	2014	10	7.82e-01	5.98e+00	2.51e+00	2.61e+00	1.16e+00	3.43e-01	5.74e+00	1.69e+00	1.99e+00	1.13e+00	7.91e-02	5.02e+00	1.14e+00	1.42e+00	1.06e+00
F72014	2014	10	7.64e-01	2.62e+00	1.10e+00	1.35e+00	5.21e-01	1.41e-01	3.66e+00	8.73e-01	1.11e+00	7.59e-01	3.56e-02	3.70e+00	5.73e-01	8.66e-01	8.83e-01
F82014	2014	10	7.31e+00	5.77e+01	2.12e+01	2.58e+01	1.51e+01	1.99e+00	3.76e+01	1.09e+01	1.25e+01	7.44e+00	9.95e-01	2.64e+01	7.96e+00	8.97e+00	4.64e+00
F92014	2014	10	5.59e+00	5.54e+01	2.06e+01	2.50e+01	1.38e+01	4.12e+00	3.79e+01	1.53e+01	1.73e+01	8.87e+00	3.11e+00	3.62e+01	1.10e+01	1.27e+01	6.23e+00
F102014	2014	10	1.61e+02	2.11e+03	6.89e+02	8.52e+02	5.26e+02	1.72e+01	1.46e+03	3.87e+02	4.69e+02	3.33e+02	7.59e+00	8.02e+02	2.57e+02	2.58e+02	1.74e+02
F112014	2014	10	1.68e+01	2.23e+03	8.77e+02	1.05e+03	5.94e+02	2.73e+01	1.81e+03	5.21e+02	6.76e+02	4.75e+02	4.00e+00	1.41e+03	3.60e+02	4.09e+02	2.61e+02
F122014	2014	10	1.57e-01	2.93e+00	1.92e+00	1.88e+00	5.90e-01	3.99e-02	1.75e+00	1.24e+00	1.13e+00	4.32e-01	3.74e-03	1.23e+00	8.55e-01	6.73e-01	3.93e-01
F132014	2014	10	2.02e-01	5.55e-01	3.27e-01	3.48e-01	8.98e-02	6.53e-02	3.80e-01	1.94e-01	1.98e-01	6.17e-02	5.92e-02	2.91e-01	1.57e-01	1.56e-01	5.97e-02
F142014	2014	10	2.29e-01	7.84e-01	3.66e-01	4.27e-01	1.57e-01	8.30e-02	6.83e-01	2.50e-01	3.36e-01	1.95e-01	1.92e-02	6.47e-01	1.55e-01	2.10e-01	1.75e-01
F152014	2014	10	1.51e+00	5.21e+00	3.55e+00	3.39e+00	8.89e-01	5.53e-01	3.84e+00	1.61e+00	1.80e+00	8.74e-01	3.26e-01	2.60e+00	1.26e+00	1.27e+00	5.38e-01
F162014	2014	10	2.43e+00	4.06e+00	3.13e+00	3.16e+00	3.24e-01	2.03e+00	3.54e+00	2.76e+00	2.74e+00	3.75e-01	1.24e+00	3.22e+00	2.28e+00	2.31e+00	4.07e-01
F172014	2014	10	2.66e+03	2.48e+05	2.53e+04	5.20e+04	6.08e+04	1.67e+03	9.73e+04	1.49e+04	1.85e+04	1.63e+04	2.42e+03	2.15e+04	9.34e+03	1.00e+04	4.97e+03
F182014	2014	10	1.00e+04	6.60e+04	3.64e+04	3.81e+04	1.38e+04	1.05e+04	5.49e+04	2.82e+04	2.99e+04	1.12e+04	5.90e+03	3.50e+04	2.01e+04	2.03e+04	7.66e+03
F192014	2014	10	5.99e+00	5.29e+02	1.52e+01	3.16e+01	7.43e+01	3.00e+00	8.33e+01	9.83e+00	1.47e+01	1.49e+01	2.08e+00	5.16e+01	6.50e+00	8.34e+00	7.11e+00
F202014	2014	10	7.78e+03	7.07e+04	2.43e+04	2.76e+04	1.49e+04	3.81e+03	7.39e+04	2.16e+04	2.36e+04	1.21e+04	5.51e+03	4.51e+04	1.57e+04	1.75e+04	8.43e+03
F212014	2014	10	9.33e+02	6.85e+05	1.09e+04	2.82e+04	9.44e+04	1.19e+03	5.59e+05	7.94e+03	1.99e+04	7.73e+04	9.28e+02	1.09e+04	5.08e+03	5.07e+03	2.50e+03
F222014	2014	10	2.36e+01	7.42e+02	2.42e+02	1.97e+02	1.61e+02	2.04e+01	3.74e+02	2.39e+02	1.64e+02	1.20e+02	2.02e+01	2.44e+02	2.37e+02	1.35e+02	1.08e+02
F232014	2014	10	2.00e+02	2.54e+02	2.42e+02	2.41e+02	6.29e+00	2.00e+02	2.42e+02	2.00e+02	2.02e+02	9.22e+00	2.00e+02	2.37e+02	2.00e+02	2.01e+02	7.22e+00
F242014	2014	10	1.13e+02	1.61e+02	1.23e+02	1.27e+02	1.30e+01	1.10e+02	1.46e+02	1.21e+02	1.21e+02	7.77e+00	1.09e+02	1.38e+02	1.17e+02	1.18e+02	4.85e+00
F252014	2014	10	2.00e+02	2.04e+02	2.02e+02	2.02e+02	8.81e-01	1.75e+02	2.04e+02	2.01e+02	2.01e+02	3.71e+00	2.00e+02	2.02e+02	2.01e+02	2.01e+02	4.24e-01
F262014	2014	10	1.00e+02	1.01e+02	1.00e+02	1.00e+02	9.29e-02	1.00e+02	1.00e+02	1.00e+02	1.00e+02	7.85e-02	1.00e+02	1.00e+02	1.00e+02	1.00e+02	7.39e-02
F272014	2014	10	5.66e+00	4.56e+02	4.02e+02	3.83e+02	5.93e+01	6.02e+00	4.33e+02	4.01e+02	3.79e+02	6.28e+01	5.71e+00	4.04e+02	4.01e+02	3.73e+02	6.25e+01
F282014	2014	10	3.70e+02	1.05e+03	4.60e+02	4.80e+02	1.17e+02	1.41e+02	6.85e+02	4.28e+02	4.39e+02	8.89e+01	2.53e+02	6.56e+02	3.79e+02	4.28e+02	7.76e+01
F292014	2014	10	4.51e+06	1.18e+07	6.15e+06	6.58e+06	1.55e+06	4.50e+06	8.49e+06	5.31e+06	5.71e+06	9.49e+05	4.50e+06	7.53e+06	5.31e+06	5.44e+06	7.19e+05
F302014	2014	10	6.73e+03	9.40e+07	3.29e+05	6.63e+06	1.86e+07	4.00e+03	2.68e+07	1.62e+05	2.18e+06	5.93e+06	3.77e+03	2.60e+07	8.57e+04	9.61e+05	3.87e+06
F12017	2017	10	2.66e+05	4.96e+08	2.24e+06	3.13e+07	9.38e+07	3.08e+03	5.01e+08	3.96e+04	1.90e+07	8.64e+07	1.22e+02	3.48e+08	6.04e+03	8.63e+06	4.90e+07
F22017	2017	10	1.56e+00	4.53e+03	9.96e+01	4.97e+02	8.12e+02	3.16e-01	4.00e+02	1.09e+01	4.90e+01	8.89e+01	5.04e-03	1.72e+03	3.21e+00	7.49e+01	2.50e+02
F32017	2017	10	3.63e+00	1.04e+02	1.65e+01	2.59e+01	2.32e+01	3.34e+00	6.96e+01	2.35e+01	2.09e+01	1.45e+01	1.49e-02	3.26e+01	9.03e+00	1.60e+01	1.12e+01
F42017	2017	10	3.68e+01	4.06e+02	7.13e+01	9.54e+01	6.37e+01	8.23e+00	3.57e+02	4.86e+01	6.02e+01	5.10e+01	4.17e+00	4.04e+02	2.83e+01	4.93e+01	7.44e+01
F52017	2017	10	1.22e-08	8.90e-03	2.90e-05	8.39e-04	1.77e-03	4.13e-11	2.11e-03	8.74e-07	9.24e-05	3.28e-04	0.00e+00	1.27e-04	1.71e-08	8.76e-06	2.64e-05
F62017	2017	10	8.06e+01	7.39e+03	2.66e+02	9.13e+02	1.34e+03	3.28e+01	6.47e+03	1.00e+02	5.15e+02	1.02e+03	7.57e+00	5.91e+03	4.82e+01	5.06e+02	9.78e+02
F72017	2017	10	1.40e+01	9.60e+01	4.51e+01	4.60e+01	1.84e+01	1.20e+01	5.07e+01	2.60e+01	2.79e+01	8.69e+00	1.02e+01	3.60e+01	2.20e+01	2.27e+01	6.36e+00
F82017	2017	10	3.50e-04	1.03e+00	8.93e-02	1.22e-01	1.96e-01	5.17e-05	1.16e+00	2.23e-02	8.19e-02	1.87e-01	8.17e-08	1.16e+00	9.06e-02	1.04e-01	1.87e-01
F92017	2017	10	1.52e+01	1.91e+03	7.52e+02	9.48e+02	5.37e+02	4.02e+01	1.53e+03	3.68e+02	4.34e+02	3.13e+02	7.61e+00	6.42e+02	1.70e+02	2.48e+02	1.88e+02
F102017	2017	10	3.09e+03	5.16e+04	1.61e+04	1.71e+04	1.02e+04	2.61e+03	3.60e+04	1.11e+04	1.30e+04	7.58e+03	4.54e+02	3.09e+04	7.47e+03	8.31e+03	5.76e+03
F112017	2017	10	1.51e+04	2.77e+07	2.97e+06	5.35e+06	6.11e+06	4.05e+03	1.25e+07	2.66e+06	2.99e+06	2.89e+06	3.26e+03	6.41e+06	2.41e+06	2.25e+06	1.85e+06
F122017	2017	10	3.71e+03	3.69e+05	1.84e+04	7.00e+04	1.05e+05	3.87e+03	1.64e+05	1.39e+04	3.06e+04	3.64e+04	4.04e+03	5.48e+04	1.10e+04	1.32e+04	9.04e+03
F132017	2017	10	5.45e+03	4.02e+04	1.48e+04	1.71e+04	8.83e+03	1.98e+02	2.28e+04	6.97e+03	7.98e+03	5.17e+03	3.41e+01	1.32e+04	1.14e+03	3.25e+03	3.37e+03
F142017	2017	10	7.06e+03	9.35e+05	1.36e+05	1.93e+05	2.09e+05	8.87e+03	3.23e+05	2.65e+04	8.15e+04	8.80e+04	5.47e+03	2.89e+05	1.52e+04	5.37e+04	7.45e+04
F152017	2017	10	1.63e+02	1.98e+04	1.47e+03	2.82e+03	4.22e+03	4.12e+00	2.22e+03	7.77e+02	7.82e+02	5.42e+02	3.07e+00	1.49e+03	4.03e+02	4.99e+02	4.44e+02
F162017	2017	10	5.99e+01	3.87e+04	4.10e+03	8.83e+03	1.07e+04	3.39e+01	2.05e+04	8.93e+02	1.93e+03	3.23e+03	2.25e+01	4.81e+03	4.81e+02	5.56e+02	7.10e+02
F172017	2017	10	5.96e+03	5.59e+04	2.49e+04	2.71e+04	1.11e+04	3.19e+03	3.91e+04	2.08e+04	1.93e+04	8.70e+03	3.26e+03	3.29e+04	1.21e+04	1.30e+04	7.66e+03
F182017	2017	10	2.56e+02	3.84e+04	1.08e+04	1.41e+04	9.59e+03	1.22e+03	3.34e+04	9.18e+03	1.09e+04	7.79e+03	2.29e+01	1.59e+04	3.19e+03	5.28e+03	4.87e+03
F192017	2017	10	3.50e+01	4.09e+02	7.74e+01	1.12e+02	8.44e+01	1.96e+01	2.64e+02	5.80e+01	8.25e+01	6.44e+01	1.30e+01	1.80e+02	3.87e+01	5.20e+01	3.55e+01

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	10	1.40e+02	2.79e+02	2.23e+02	2.19e+02	3.50e+01	1.04e+02	2.85e+02	2.02e+02	1.94e+02	3.86e+01	1.01e+02	2.66e+02	1.87e+02	1.76e+02	4.14e+01
F212017	2017	10	1.01e+02	1.11e+02	1.05e+02	1.05e+02	2.72e+00	1.00e+02	1.10e+02	1.04e+02	1.05e+02	2.92e+00	1.00e+02	1.10e+02	1.04e+02	1.04e+02	2.27e+00
F222017	2017	10	1.24e+02	5.25e+02	2.41e+02	2.55e+02	9.46e+01	1.02e+02	5.99e+02	2.03e+02	2.11e+02	8.99e+01	1.00e+02	5.73e+02	2.00e+02	2.00e+02	8.82e+01
F232017	2017	10	2.34e+02	8.92e+02	4.29e+02	4.84e+02	1.66e+02	2.10e+02	6.74e+02	2.76e+02	3.58e+02	1.67e+02	2.06e+02	6.56e+02	2.48e+02	3.42e+02	1.79e+02
F242017	2017	10	4.79e+02	5.86e+02	4.91e+02	4.99e+02	2.26e+01	4.77e+02	5.19e+02	4.86e+02	4.91e+02	1.19e+01	4.76e+02	5.18e+02	4.84e+02	4.90e+02	1.25e+01
F252017	2017	10	4.58e+02	4.89e+02	4.75e+02	4.74e+02	7.03e+00	4.58e+02	4.86e+02	4.67e+02	4.67e+02	5.95e+00	4.57e+02	4.86e+02	4.67e+02	4.67e+02	4.89e+00
F262017	2017	10	3.91e+02	4.80e+02	3.96e+02	4.03e+02	1.89e+01	3.90e+02	4.76e+02	3.94e+02	3.95e+02	1.22e+01	3.88e+02	4.34e+02	3.91e+02	3.94e+02	8.04e+00
F272017	2017	10	2.60e+01	4.77e+02	1.19e+02	1.71e+02	1.55e+02	9.18e-01	4.37e+02	1.05e+01	9.86e+01	1.39e+02	5.79e-02	4.38e+02	2.21e-01	5.77e+01	1.24e+02
F282017	2017	10	1.62e+03	1.59e+06	1.93e+04	6.30e+04	2.21e+05	1.51e+03	6.57e+04	8.06e+03	1.32e+04	1.34e+04	1.45e+03	2.52e+04	4.73e+03	5.97e+03	5.42e+03
F292017	2017	10	1.39e+04	4.79e+07	5.34e+04	1.44e+06	6.91e+06	3.13e+03	4.64e+05	2.79e+04	5.17e+04	7.46e+04	4.08e+03	1.47e+05	2.56e+04	3.07e+04	2.70e+04
F12022	2022	10	2.86e+00	1.69e+03	8.63e+01	1.52e+02	2.96e+02	5.00e+00	2.19e+02	6.39e+01	7.09e+01	4.77e+01	4.47e-03	1.52e+02	4.24e+01	4.36e+01	3.69e+01
F22022	2022	10	6.18e+00	7.75e+01	1.32e+01	2.78e+01	2.29e+01	4.88e-01	7.18e+01	1.12e+01	2.16e+01	2.07e+01	1.25e-01	7.23e+01	1.08e+01	1.72e+01	1.97e+01
F32022	2022	10	8.50e-06	2.63e-02	2.74e-04	1.33e-03	3.78e-03	5.08e-08	1.32e-02	8.40e-05	7.63e-04	1.92e-03	3.36e-09	1.36e-02	1.47e-06	7.64e-04	1.96e-03
F42022	2022	10	1.40e+01	1.06e+02	4.56e+01	4.88e+01	1.98e+01	1.20e+01	5.85e+01	3.00e+01	3.00e+01	1.07e+01	1.00e+01	3.60e+01	2.12e+01	2.18e+01	6.33e+00
F52022	2022	10	1.24e-04	5.37e-01	1.60e-02	7.91e-02	1.35e-01	1.39e-04	5.36e-01	5.31e-03	5.76e-02	1.37e-01	4.73e-06	5.64e-01	2.60e-03	5.45e-02	1.54e-01
F62022	2022	10	7.82e+03	9.36e+04	3.41e+04	3.82e+04	1.87e+04	7.45e+03	7.53e+04	3.26e+04	3.44e+04	1.66e+04	7.35e+03	6.34e+04	3.11e+04	3.41e+04	1.66e+04
F72022	2022	10	2.63e+01	1.98e+02	4.52e+01	6.85e+01	4.59e+01	2.28e+01	1.41e+02	3.16e+01	4.16e+01	3.08e+01	1.04e+01	1.40e+02	2.49e+01	2.93e+01	2.16e+01
F82022	2022	10	3.03e+01	2.95e+03	5.61e+02	8.09e+02	7.72e+02	2.27e+01	1.85e+03	4.73e+01	2.34e+02	5.15e+02	2.64e+00	1.37e+02	2.44e+01	2.94e+01	1.89e+01
F92022	2022	10	4.64e+00	4.99e+02	3.74e+02	3.10e+02	1.71e+02	3.12e-01	4.89e+02	3.66e+02	2.79e+02	1.73e+02	2.31e-02	4.15e+02	2.06e+02	1.86e+02	1.84e+02
F102022	2022	10	2.21e+02	3.69e+02	2.24e+02	2.52e+02	4.92e+01	2.21e+02	3.59e+02	2.23e+02	2.51e+02	4.82e+01	2.21e+02	3.34e+02	2.22e+02	2.49e+02	4.44e+01
F112022	2022	10	1.99e-01	8.79e+02	3.44e+00	3.85e+01	1.71e+02	7.60e-02	8.77e+02	3.20e+00	3.78e+01	1.71e+02	2.93e-03	8.76e+02	3.60e-02	3.61e+01	1.71e+02
F122022	2022	10	1.64e+02	2.13e+02	1.67e+02	1.70e+02	9.22e+00	1.64e+02	1.82e+02	1.65e+02	1.66e+02	3.07e+00	1.64e+02	1.81e+02	1.65e+02	1.66e+02	2.75e+00

GWO — estadisticas por funcion, D = 50

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	50	7.48e+07	4.06e+08	2.04e+08	2.02e+08	7.07e+07	3.20e+07	2.89e+08	9.69e+07	1.07e+08	5.37e+07	2.49e+07	1.94e+08	6.37e+07	7.45e+07	3.58e+07
F22014	2014	50	3.07e+09	1.65e+10	8.40e+09	8.38e+09	3.18e+09	9.05e+08	1.46e+10	6.73e+09	6.97e+09	2.75e+09	7.83e+08	1.55e+10	6.32e+09	6.35e+09	2.92e+09
F32014	2014	50	1.46e+05	3.46e+05	2.11e+05	2.18e+05	4.76e+04	4.33e+04	1.61e+05	9.79e+04	1.01e+05	2.25e+04	2.65e+04	9.03e+04	5.81e+04	5.89e+04	1.32e+04
F42014	2014	50	3.89e+02	3.00e+03	1.09e+03	1.20e+03	5.51e+02	2.34e+02	1.94e+03	6.36e+02	7.15e+02	3.42e+02	2.41e+02	1.38e+03	5.36e+02	5.87e+02	2.48e+02
F52014	2014	50	2.12e+01	2.14e+01	2.13e+01	2.13e+01	4.20e-02	2.11e+01	2.13e+01	2.12e+01	2.12e+01	3.07e-02	2.11e+01	2.12e+01	2.11e+01	2.11e+01	2.68e-02
F62014	2014	50	2.98e+01	4.97e+01	3.89e+01	3.89e+01	4.08e+00	2.48e+01	4.36e+01	3.37e+01	3.36e+01	4.47e+00	2.21e+01	3.95e+01	2.98e+01	3.04e+01	4.33e+00
F72014	2014	50	2.16e+01	1.80e+02	8.09e+01	8.31e+01	3.69e+01	1.70e+01	1.72e+02	4.72e+01	5.45e+01	2.83e+01	1.24e+01	1.03e+02	4.08e+01	4.51e+01	2.40e+01
F82014	2014	50	1.55e+02	5.02e+02	2.51e+02	2.75e+02	8.75e+01	1.25e+02	3.98e+02	1.88e+02	1.97e+02	5.82e+01	1.16e+02	3.83e+02	1.85e+02	1.87e+02	3.89e+01
F92014	2014	50	1.88e+02	5.42e+02	2.84e+02	3.09e+02	8.39e+01	1.64e+02	5.35e+02	2.14e+02	2.32e+02	7.95e+01	1.33e+02	4.46e+02	1.83e+02	1.96e+02	5.82e+01
F102014	2014	50	5.67e+03	1.57e+04	8.05e+03	9.20e+03	2.99e+03	4.22e+03	1.42e+04	5.92e+03	6.52e+03	2.14e+03	3.64e+03	1.14e+04	5.55e+03	5.71e+03	1.31e+03
F112014	2014	50	6.76e+03	1.57e+04	1.45e+04	1.27e+04	2.93e+03	3.33e+03	1.46e+04	7.72e+03	9.65e+03	3.89e+03	3.61e+03	1.37e+04	5.80e+03	6.38e+03	2.09e+03
F122014	2014	50	3.50e+00	5.72e+00	4.85e+00	4.75e+00	4.92e-01	2.77e-01	4.65e+00	4.05e+00	3.77e+00	1.01e+00	6.86e-02	3.85e+00	3.35e+00	3.01e+00	1.09e+00
F132014	2014	50	6.37e-01	2.78e+00	8.72e-01	9.96e-01	4.67e-01	5.63e-01	2.23e+00	7.74e-01	8.38e-01	3.32e-01	4.97e-01	8.54e-01	6.44e-01	6.53e-01	7.46e-02
F142014	2014	50	4.41e-01	5.60e+01	1.93e+01	1.87e+01	1.30e+01	3.06e-01	3.12e+01	1.00e+01	1.17e+01	9.98e+00	2.37e-01	3.55e+01	4.36e+00	1.06e+01	1.15e+01
F152014	2014	50	7.39e+01	2.19e+04	2.17e+03	3.70e+03	4.90e+03	5.83e+01	5.77e+03	4.38e+02	9.44e+02	1.26e+03	2.34e+01	2.17e+04	2.50e+02	1.51e+03	3.76e+03
F162014	2014	50	2.16e+01	2.38e+01	2.28e+01	2.28e+01	4.95e-01	1.95e+01	2.34e+01	2.19e+01	2.18e+01	7.56e-01	1.88e+01	2.25e+01	2.12e+01	2.10e+01	7.42e-01
F172014	2014	50	1.45e+06	4.85e+07	1.29e+07	1.44e+07	1.03e+07	7.80e+05	2.78e+07	3.52e+06	4.93e+06	4.41e+06	3.24e+05	1.04e+07	1.76e+06	2.41e+06	2.04e+06
F182014	2014	50	3.43e+05	9.60e+07	1.83e+06	4.90e+06	1.34e+07	1.10e+05	2.31e+06	1.90e+05	3.45e+05	4.31e+05	1.01e+05	1.77e+07	1.63e+05	6.30e+05	2.49e+06
F192014	2014	50	3.59e+02	6.09e+05	8.21e+03	5.22e+04	1.06e+05	6.16e+01	1.54e+05	1.25e+03	1.16e+04	3.01e+04	4.05e+01	3.58e+04	2.37e+02	4.09e+03	7.38e+03
F202014	2014	50	1.81e+05	8.68e+09	8.84e+06	7.34e+08	1.85e+09	1.47e+05	2.23e+09	2.32e+05	7.62e+07	3.39e+08	1.07e+05	2.06e+09	2.23e+05	7.39e+07	3.41e+08
F212014	2014	50	6.20e+05	1.28e+08	1.85e+07	2.47e+07	2.22e+07	7.85e+05	4.19e+07	4.44e+06	8.91e+06	1.08e+07	2.74e+05	4.62e+07	1.99e+06	3.90e+06	6.91e+06
F222014	2014	50	2.97e+03	1.44e+06	5.25e+03	3.85e+04	2.03e+05	2.74e+03	3.55e+06	4.20e+03	9.08e+04	5.09e+05	1.14e+03	4.86e+03	2.97e+03	2.97e+03	8.32e+02
F232014	2014	50	2.37e+02	2.45e+02	2.38e+02	2.39e+02	2.06e+00	2.36e+02	2.39e+02	2.36e+02	2.37e+02	6.88e-01	2.36e+02	2.39e+02	2.36e+02	2.36e+02	4.34e-01
F242014	2014	50	2.40e+02	3.03e+02	2.71e+02	2.68e+02	1.27e+01	2.00e+02	2.00e+02	2.00e+02	2.00e+02	3.76e-02	2.00e+02	2.00e+02	2.00e+02	2.00e+02	3.42e-03
F252014	2014	50	2.36e+02	2.66e+02	2.49e+02	2.49e+02	7.45e+00	2.00e+02	2.48e+02	2.37e+02	2.34e+02	1.07e+01	2.00e+02	2.44e+02	2.30e+02	2.31e+02	8.11e+00
F262014	2014	50	1.01e+02	3.96e+02	2.03e+02	2.04e+02	6.56e+01	1.01e+02	3.84e+02	2.00e+02	1.96e+02	7.22e+01	1.01e+02	3.70e+02	2.00e+02	1.86e+02	6.83e+01
F272014	2014	50	1.09e+03	1.46e+03	1.28e+03	1.28e+03	8.46e+01	7.69e+02	1.41e+03	1.10e+03	1.09e+03	1.60e+02	7.22e+02	1.32e+03	1.02e+03	1.01e+03	1.11e+02
F282014	2014	50	2.32e+03	8.04e+03	4.15e+03	4.25e+03	1.08e+03	1.59e+03	4.32e+03	2.45e+03	2.63e+03	7.22e+02	1.47e+03	3.39e+03	2.15e+03	2.20e+03	5.17e+02
F292014	2014	50	8.35e+06	4.50e+08	1.40e+08	1.89e+08	1.40e+08	6.75e+05	4.31e+08	4.83e+07	7.18e+07	8.13e+07	3.73e+05	2.05e+08	2.38e+07	4.90e+07	5.33e+07
F302014	2014	50	1.77e+08	2.26e+09	8.46e+08	9.09e+08	4.83e+08	2.75e+07	1.78e+09	3.39e+08	3.76e+08	2.90e+08	1.73e+07	7.97e+08	2.05e+08	2.52e+08	1.85e+08
F12017	2017	50	2.89e+09	1.75e+10	8.48e+09	9.04e+09	3.74e+09	1.64e+09	1.41e+10	5.60e+09	6.10e+09	2.87e+09	1.83e+09	1.04e+10	5.09e+09	5.13e+09	1.84e+09
F22017	2017	50	2.64e+04	7.88e+04	5.00e+04	5.04e+04	1.12e+04	7.90e+03	4.56e+04	2.81e+04	2.76e+04	9.43e+03	5.49e+03	4.38e+04	2.29e+04	2.35e+04	7.46e+03
F32017	2017	50	2.89e+02	2.24e+03	7.60e+02	8.84e+02	4.67e+02	2.46e+02	1.64e+03	3.82e+02	4.71e+02	2.59e+02	1.92e+02	1.35e+03	3.11e+02	3.98e+02	2.34e+02
F42017	2017	50	2.77e+03	1.74e+04	7.11e+03	7.92e+03	3.61e+03	1.58e+03	1.28e+04	4.47e+03	5.57e+03	2.99e+03	1.64e+03	1.43e+04	4.47e+03	5.05e+03	2.48e+03
F52017	2017	50	4.74e-04	5.43e-02	2.29e-03	6.82e-03	1.18e-02	3.59e-05	6.88e-03	5.86e-04	9.37e-04	1.24e-03	5.70e-05	2.03e-03	3.48e-04	4.31e-04	3.61e-04
F62017	2017	50	9.26e+04	6.89e+05	3.17e+05	3.46e+05	1.36e+05	5.98e+04	6.32e+05	2.33e+05	2.65e+05	1.39e+05	3.58e+04	4.14e+05	2.16e+05	2.17e+05	8.49e+04
F72017	2017	50	3.25e+02	1.04e+03	5.88e+02	6.47e+02	2.16e+02	2.76e+02	9.23e+02	4.01e+02	4.45e+02	1.53e+02	2.46e+02	8.56e+02	3.67e+02	4.14e+02	1.49e+02
F82017	2017	50	3.72e+00	2.86e+01	1.23e+01	1.28e+01	4.62e+00	3.04e+00	1.56e+01	8.62e+00	8.78e+00	3.38e+00	2.92e+00	1.76e+01	7.57e+00	7.40e+00	2.92e+00
F92017	2017	50	7.98e+03	1.60e+04	1.13e+04	1.20e+04	2.94e+03	4.60e+03	1.51e+04	7.17e+03	9.01e+03	3.49e+03	4.12e+03	1.39e+04	6.52e+03	7.42e+03	2.74e+03
F102017	2017	50	1.68e+05	5.15e+07	2.91e+06	9.49e+06	1.37e+07	1.14e+05	3.31e+07	5.66e+05	3.12e+06	7.26e+06	9.01e+04	4.27e+07	4.93e+05	4.03e+06	9.38e+06
F112017	2017	50	3.73e+07	1.27e+09	2.38e+08	3.66e+08	3.20e+08	1.03e+07	1.45e+09	1.65e+08	2.41e+08	2.52e+08	9.34e+06	6.56e+08	4.11e+07	1.13e+08	1.59e+08
F122017	2017	50	8.10e+07	3.01e+09	4.77e+08	6.38e+08	5.50e+08	8.39e+06	1.70e+09	2.25e+08	3.23e+08	3.24e+08	2.61e+06	2.37e+09	1.16e+08	2.87e+08	3.91e+08
F132017	2017	50	4.94e+05	8.44e+06	2.52e+06	2.91e+06	1.86e+06	2.67e+05	3.95e+06	9.07e+05	1.11e+06	7.39e+05	2.21e+05	1.73e+06	5.00e+05	6.24e+05	3.63e+05
F142017	2017	50	3.88e+06	1.42e+09	3.15e+07	1.20e+08	2.43e+08	1.10e+06	8.60e+08	9.85e+06	6.85e+07	1.57e+08	1.37e+05	3.52e+08	4.12e+06	2.78e+07	7.61e+07
F152017	2017	50	5.86e+04	5.01e+07	1.31e+06	5.97e+06	1.13e+07	3.67e+04	4.74e+07	6.68e+05	3.39e+06	9.10e+06	3.53e+04	2.48e+07	1.02e+05	2.02e+06	4.63e+06
F162017	2017	50	5.09e+04	1.12e+05	7.26e+04	7.48e+04	1.63e+04	2.09e+04	7.58e+04	4.88e+04	4.81e+04	1.13e+04	1.44e+04	7.75e+04	3.10e+04	3.16e+04	9.60e+03
F172017	2017	50	9.33e+05	2.92e+07	7.04e+06	8.91e+06	6.70e+06	4.56e+05	1.82e+07	2.84e+06	4.29e+06	3.84e+06	2.90e+05	2.32e+07	1.79e+06	2.65e+06	3.51e+06
F182017	2017	50	8.37e+09	8.81e+10	2.95e+10	3.39e+10	1.85e+10	6.28e+08	5.50e+10	1.24e+10	1.68e+10	1.37e+10	1.26e+07	3.32e+10	8.35e+09	1.03e+10	8.27e+09
F192017	2017	50	5.81e+02	3.34e+03	1.80e+03	1.83e+03	5.92e+02	2.46e+02	3.14e+03	9.57e+02	1.10e+03	5.80e+02	2.27e+02	2.01e+03	8.27e+02	8.62e+02	4.46e+02

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	50	1.39e+03	2.28e+04	8.22e+03	8.45e+03	5.11e+03	1.34e+03	1.57e+04	6.12e+03	6.31e+03	3.51e+03	7.39e+02	1.99e+04	4.79e+03	5.30e+03	3.56e+03
F212017	2017	50	2.85e+02	8.79e+03	3.53e+02	5.38e+02	1.18e+03	2.42e+02	1.67e+03	2.98e+02	3.31e+02	1.95e+02	2.01e+02	4.79e+02	2.77e+02	2.89e+02	4.56e+01
F222017	2017	50	5.18e+03	3.84e+04	1.76e+04	1.98e+04	8.32e+03	5.06e+03	2.71e+04	1.26e+04	1.27e+04	4.78e+03	4.07e+03	1.99e+04	1.12e+04	1.10e+04	3.36e+03
F232017	2017	50	3.40e+03	2.92e+04	8.67e+03	9.63e+03	4.30e+03	2.58e+03	1.28e+04	6.63e+03	6.73e+03	2.03e+03	1.02e+03	9.87e+03	6.22e+03	6.29e+03	2.05e+03
F242017	2017	50	1.40e+03	3.82e+03	2.14e+03	2.20e+03	5.04e+02	1.26e+03	2.90e+03	1.61e+03	1.76e+03	4.29e+02	9.67e+02	2.35e+03	1.58e+03	1.62e+03	3.47e+02
F252017	2017	50	1.80e+03	3.31e+03	2.23e+03	2.29e+03	3.40e+02	1.67e+03	2.86e+03	1.86e+03	1.91e+03	2.22e+02	1.56e+03	2.48e+03	1.84e+03	1.88e+03	1.71e+02
F262017	2017	50	6.85e+02	1.15e+03	8.52e+02	8.64e+02	1.11e+02	5.93e+02	1.03e+03	7.12e+02	7.35e+02	8.60e+01	5.88e+02	8.76e+02	6.70e+02	6.77e+02	5.81e+01
F272017	2017	50	4.92e+02	8.65e+02	5.63e+02	5.79e+02	7.10e+01	4.84e+02	6.65e+02	5.22e+02	5.30e+02	3.87e+01	4.75e+02	6.64e+02	5.13e+02	5.19e+02	3.21e+01
F282017	2017	50	1.55e+08	5.30e+09	1.55e+09	1.65e+09	1.07e+09	3.62e+07	1.79e+09	3.94e+08	5.00e+08	3.82e+08	2.62e+07	1.15e+09	2.45e+08	2.76e+08	2.09e+08
F292017	2017	50	1.13e+09	1.80e+10	8.61e+09	9.05e+09	4.14e+09	6.63e+08	1.24e+10	4.25e+09	4.88e+09	3.04e+09	1.97e+08	1.19e+10	2.18e+09	3.06e+09	2.49e+09
F12022	2022	20	5.79e+01	1.06e+04	1.99e+03	2.83e+03	2.44e+03	6.54e+01	4.52e+03	7.73e+02	1.10e+03	1.06e+03	3.78e+01	4.12e+03	5.10e+02	9.02e+02	9.81e+02
F22022	2022	20	5.12e+01	2.13e+02	9.09e+01	1.02e+02	3.78e+01	4.81e+01	1.63e+02	7.80e+01	8.34e+01	2.30e+01	4.61e+01	1.69e+02	7.75e+01	8.06e+01	2.53e+01
F32022	2022	20	4.13e-04	5.74e-02	1.74e-02	2.01e-02	1.39e-02	1.66e-05	5.31e-02	6.42e-03	1.11e-02	1.31e-02	5.75e-07	3.80e-02	7.30e-03	1.11e-02	1.12e-02
F42022	2022	20	4.61e+01	2.91e+02	1.65e+02	1.59e+02	7.65e+01	4.40e+01	2.16e+02	1.14e+02	1.19e+02	5.33e+01	4.60e+01	1.70e+02	9.40e+01	9.74e+01	3.22e+01
F52022	2022	20	5.67e-02	3.64e+00	7.64e-01	1.01e+00	8.98e-01	1.70e-03	3.22e+00	4.28e-01	6.71e-01	6.40e-01	8.72e-04	1.54e+00	6.47e-01	5.94e-01	3.73e-01
F62022	2022	20	4.27e+04	3.42e+07	1.82e+05	1.57e+06	5.11e+06	1.56e+04	2.37e+07	5.99e+04	6.22e+05	3.36e+06	2.40e+04	2.19e+06	5.77e+04	9.93e+04	2.99e+05
F72022	2022	20	4.99e+01	8.23e+02	1.22e+02	1.56e+02	1.32e+02	2.79e+01	2.27e+02	5.94e+01	6.76e+01	3.88e+01	2.20e+01	1.64e+02	4.14e+01	5.00e+01	2.74e+01
F82022	2022	20	1.18e+03	1.39e+04	4.02e+03	4.14e+03	2.10e+03	7.01e+02	5.05e+03	1.96e+03	2.05e+03	9.26e+02	3.07e+02	3.58e+03	1.09e+03	1.20e+03	6.30e+02
F92022	2022	20	8.94e+01	5.32e+02	4.17e+02	4.18e+02	6.22e+01	3.40e+02	4.77e+02	3.72e+02	3.84e+02	3.84e+01	3.38e+02	4.48e+02	3.70e+02	3.77e+02	3.02e+01
F102022	2022	20	4.04e+02	5.44e+03	2.21e+03	2.24e+03	1.79e+03	4.01e+02	4.57e+03	1.32e+03	1.55e+03	1.28e+03	3.06e+02	4.12e+03	5.38e+02	1.06e+03	9.36e+02
F112022	2022	20	6.00e+00	1.36e+03	2.26e+01	6.06e+01	2.03e+02	3.45e+00	2.78e+01	1.32e+01	1.53e+01	5.51e+00	6.39e-01	4.55e+01	1.24e+01	1.38e+01	7.93e+00
F122022	2022	20	2.44e+02	3.55e+02	2.92e+02	2.92e+02	2.70e+01	2.43e+02	3.19e+02	2.62e+02	2.66e+02	1.71e+01	2.38e+02	3.05e+02	2.58e+02	2.60e+02	1.53e+01